

Applied-Bioinformatics-Structural-Analysis- **Internship**

Batch No: 03

SUMMARY:

During this internship, a comprehensive bioinformatics workflow was carried out, covering data retrieval, sequence analysis, protein characterization, structure prediction, and structural validation. The work began with retrieving protein sequences and structural data from publicly available biological databases, including NCBI, UniProt, and the Protein Data Bank (PDB). These databases served as reliable sources for obtaining the sequence and structural information required for subsequent analyses.

The retrieved protein sequences were then subjected to sequence analysis. BLAST was used to identify homologous sequences and evaluate sequence similarity, followed by Multiple Sequence Alignment (MSA) using Clustal W to identify conserved regions among the selected proteins. The aligned sequences were further analyzed through phylogenetic analysis using MEGA X, and the resulting phylogenetic trees were visualized and annotated using iTOL to study evolutionary relationships.

To characterize the proteins, physicochemical property analysis was performed using ExPASy ProtParam. Parameters such as molecular weight, theoretical isoelectric point (pI), instability index, aliphatic index, and GRAVY score were calculated to assess the biochemical properties and stability of the proteins.

Protein structure prediction was then carried out by analyzing the primary, secondary, and tertiary structural features of the selected proteins. Primary structure analysis involved evaluating the amino acid sequence and its physicochemical characteristics, while PSIPRED was used to predict the secondary structure by identifying α -helices, β -strands, and random coils. Tertiary protein structures were subsequently obtained and prepared for further refinement and quality assessment.

The predicted three-dimensional protein structures were refined using GalaxyRefine to improve their structural quality. The refined models were then validated using multiple structure validation tools, including PROCHECK for stereochemical assessment, ERRAT for evaluating non-bonded atomic interactions, and Verify3D for assessing sequence–structure compatibility. These validation steps ensured that the refined protein models were reliable and suitable for further structural and functional studies.

Overall, this internship provided practical experience with widely used biological databases and bioinformatics tools, strengthening skills in sequence retrieval, comparative sequence analysis, protein characterization, phylogenetic analysis, protein structure prediction, refinement, and validation. The workflow demonstrated how computational approaches can be integrated to investigate protein structure and function, providing a strong foundation for future research in bioinformatics and computational biology.
